Rare Disease Variant Report — Master Report

DM1 · MERRF · DMD · Haemophilia A | Generated 2026-07-15 09:31 | Pipeline v1.1 | Tool chain: bcftools, VEP, SnpEff, ClinVar/SnpSift, gnomAD, SpliceAI, REVEL, AlphaMissense, CADD, ClinGen dosage, InterVar, ClassifyCNV, SvAnna (AnnotSV-equivalent), ExpansionHunter/TRGT, pbsv/Sniffles2/Truvari

Contents

1. Executive Summary

- **DM1:** A pathogenic CTG repeat expansion of 80 repeats was detected at the DMPK locus (chr19:45,770,205), well above the ACMG/AMP pathogenic threshold of ≥50 repeats and clearly outside the normal range (≤34 repeats).
- **MERRF:** The classic MERRF hotspot m.
- **DMD:** Two independent, clinically significant findings were identified in DMD: an in-frame exon 45–47 deletion (predicted to produce a milder, Becker-like phenotype because the reading frame is preserved) and a separate nonsense SNV (stop_gained, predicted frameshift/truncating — the classically severe, Duchenne-like mechanism).
- **HEMA_A:** Four findings were returned for F8: (1) a likely-pathogenic missense variant supported concordantly by REVEL, AlphaMissense, and CADD as well as ClinVar/InterVar; (2) a large exon 1–2 deletion, pathogenic by both ClassifyCNV and SvAnna and consistent with F8 haploinsufficiency; (3) a candidate intron 22 inversion breakend — the classic Haemophilia A inversion hotspot — that is flagged for manual confirmation by long-read sequencing or inverse-PCR rather than auto-called, since only a single tool (SV caller) reports it; and (4) a common benign intronic variant.

13 variants were returned across the four disease regions. 8 are classified Pathogenic or Likely Pathogenic, 1 is a flagged structural-variant candidate awaiting orthogonal confirmation, and the remainder are benign/common variants retained for transparency. One discordant call (MERRF m.8356T>C) requires manual adjudication — see Section 4.

2. Tool Glossary

Every tool in the chain and what it contributes to each row below:

Tool	What it does / what it contributes
bcftools	Variant-calling and VCF QC/normalization utility. Produces per-sample QUAL/DP distributions and PASS rates used in the Quality Control step before any annotation happens.
VEP (Ensembl Variant Effect Predictor)	Predicts the functional consequence of each variant on transcripts (e.g. missense_variant, stop_gained, intron_variant) and assigns an IMPACT tier (HIGH/MODERATE/LOW/MODIFIER).
SnpEff	A second, independent functional-consequence annotator run alongside VEP so consequence calls can be cross-checked (columns vep_consequence vs snpeff_consequence).
ClinVar / SnpSift	Looks up each variant against the public ClinVar database of clinically reported variants, returning a clinical significance label (Pathogenic, Likely_pathogenic, Uncertain_significance, Likely_benign, Benign, or not_in_clinvar).
gnomAD	Population allele-frequency database. A variant that is common in the general population (gnomad_af above the 'common' threshold, 0.01 in this pipeline) is penalized in scoring; a rare variant (AF ≤ 0.001) is rewarded.
SpliceAI	Deep-learning splicing predictor. spliceai_delta is the maximum predicted change in splice-site usage; a delta ≥0.5 is treated as evidence of splice disruption.
REVEL	Ensemble missense-pathogenicity predictor (0–1 score). A score ≥0.7 is treated as evidence the missense change is damaging.
AlphaMissense	DeepMind's missense-pathogenicity predictor (0–1 score). A score ≥0.564 is treated as evidence of a pathogenic effect.
CADD	Combined Annotation Dependent Depletion — a general deleteriousness score (phred-scaled). A score ≥20 (top ~1% most deleterious substitutions genome-wide) counts as supporting evidence.
ClinGen Dosage Sensitivity	Curated gene-level statement of whether a gene is known to be haploinsufficient (loss of one copy causes disease). Used to support interpretation of deletions/CNVs (e.g. DMD and F8 are both HI=3, 'sufficient evidence').
InterVar	Automates the ACMG/AMP 2015 sequence-variant classification guidelines (PVS1, PS1–4, PM1–6, PP1–5, BA1, BS1–4, BP1–7 evidence codes) into a five-tier call: Pathogenic, Likely Pathogenic, Uncertain Significance, Likely Benign, Benign. This pipeline also computes a lightweight, complementary point-based version of the same rules directly (Section 9).
ClassifyCNV	ACMG/ClinGen-based classifier specifically for copy-number variants (deletions/duplications) — assigns pathogenicity to CNVs like the DMD exon 45-47 deletion and the F8 exon 1-2 deletion.
SvAnna (standing in for AnnotSV)	Structural-variant functional annotator/prioritizer. Used here in place of AnnotSV to annotate SV consequence (e.g. loss of start codon, in-frame vs out-of-frame exon loss) and to help interpret breakends.
ExpansionHunter / TRGT	Repeat-expansion genotypers. ExpansionHunter sizes short tandem repeats from short-read BAMs (used for the DMPK CTG repeat); TRGT is the equivalent long-read (PacBio HiFi) genotyper, included for higher-confidence sizing when long reads are available.
pbsv / Sniffles2 / Truvari	Long-read structural-variant callers (pbsv, Sniffles2) used to detect exon-level deletions and candidate inversions (e.g. DMD exon deletions, the F8 intron 22 inversion breakend); Truvari is used to merge/benchmark SV calls across callers.

3. Column Glossary

How to read every column that appears in the variant tables below:

Column(s)	Meaning
gene / chrom / pos / ref / alt	Standard variant identifiers: gene symbol, chromosome, 1-based position (GRCh38), reference and alternate allele. Structural variants use symbolic ALT alleles (~~,) and repeat expansions use a synthetic allele like (80 repeat units).~~
variant_type	Pipeline-assigned category: point_mutation, missense, nonsense, synonymous, intron_variant, exon_deletion, candidate_inversion, or repeat_expansion.
impact	VEP/SnpEff-derived severity tier: HIGH (e.g. stop_gained, transcript_ablation), MODERATE (e.g. missense), LOW (e.g. synonymous), MODIFIER (e.g. intronic).
gnomad_af	Population allele frequency in gnomAD. Lower = rarer = more likely disease-relevant for a dominant/rare disorder.
revel / alphamissense / cadd_phred / spliceai_delta	In-silico pathogenicity prediction scores (see Tool Glossary above for each). Only populated for point mutations/missense variants — not applicable to structural variants or repeat expansions (shown as –).
clinvar_significance	Clinical significance as curated in ClinVar.
vep_consequence / snpeff_consequence	Predicted molecular consequence from two independent annotators, shown side-by-side for cross-checking.
intervar_class	ACMG/AMP classification with the specific evidence codes invoked (e.g. PVS1, PM2), from InterVar / the pipeline's rule engine.
sv_cnv_class	Structural-variant / CNV specific classification from ClassifyCNV and SvAnna (shown together), or the SV caller's flag for unresolved events like candidate inversions.
clingen_dosage	Gene-level haploinsufficiency/triplosensitivity evidence from ClinGen, used to support deletion/duplication interpretation.
heteroplasmy	Fraction of mitochondrial genomes carrying the variant (0–1). Only applicable to mitochondrial (MERRF) variants.
deleted_exons / frame_effect	For exon-level CNVs: which exons are deleted, and whether the deletion preserves (in-frame) or disrupts (frameshift) the reading frame — a major driver of predicted disease severity (Becker-like vs Duchenne-like).
priority_score / score_reasons	The pipeline's own weighted score (Section 8) combining gene match, functional impact, population frequency, and in-silico predictors, with a human-readable breakdown of exactly which rules fired.
tool_agreement / n_tools_reporting	Whether the independent classification tools (ClinVar, InterVar, ClassifyCNV/ SvAnna) agree on pathogenicity for this variant, and how many of them returned a call at all. Discordant calls are highlighted and should be manually reviewed.

4. Myotonic Dystrophy Type 1 (DMPK, CTG repeat expansion)

> A pathogenic CTG repeat expansion of 80 repeats was detected at the DMPK locus (chr19:45,770,205), well above the ACMG/AMP pathogenic threshold of ≥50 repeats and clearly outside the normal range (≤34 repeats). This is sized by

ExpansionHunter from short-read data and is reported as Pathogenic by both the repeat-expansion rule engine and ClinVar. One additional missense variant of uncertain significance and one common benign intronic variant were also found in the DMPK region but do not change the overall interpretation.

Prioritized variants — DM1

Disease	Gene	Chr	Position	Ref	Alt	Variant Type	Impact	gnomAD AF	REVEL	AlphaMissense	CADD (phred)	S Δ
DM1	DMPK	chr19	45770205	C		repeat_expansion	HIGH	–	–	–	–	–
DM1	DMPK	chr19	45771320	G	A	missense	MODERATE	0.00090	0.42	0.31	19.1	0
DM1	DMPK	chr19	45778990	T	C	intron_variant	MODIFIER	0.18000	–	–	4.2	0

Multi-tool comparison — DM1

Disease	Gene	Chr	Position	Ref	Alt	ClinVar	InterVar (ACMG)	ClassifyCNV / SvAnna	Tool Agreement	# Tools Reporting
DM1	DMPK	chr19	45770205	C		Pathogenic	Pathogenic (repeat-expansion rule)	Pathogenic (ExpansionHunter: 80 CTG repeats, threshold >=50)	Concordant	3
DM1	DMPK	chr19	45771320	G	A	Uncertain_significance	Uncertain Significance (PM2,BP4)	–	Concordant	2
DM1	DMPK	chr19	45778990	T	C	Benign	Benign (BA1)	–	Concordant	2

4. MERRF Syndrome (MT-TK, mitochondrial point mutation / heteroplasmy)

The classic MERRF hotspot m.8344A>G was detected at 78% heteroplasmy, well above the 10% pathogenic heteroplasmy threshold used by this pipeline, and is called Pathogenic by both ClinVar and InterVar (concordant). A second candidate hotspot, m.8356T>C, was also detected but at only 6% heteroplasmy — ClinVar

> lists this as Uncertain Significance while InterVar's rule engine leans Likely Pathogenic based on the known hotspot position, producing a flagged discordant call that warrants manual review (see Multi-Tool Comparison table, highlighted row). A third variant at this locus is a common, benign haplogroup marker.

Prioritized variants — MERRF

Disease	Gene	Chr	Position	Ref	Alt	Variant Type	Impact	gnomAD AF	REVEL	AlphaMissense	CADD (phred)	Splice Δ
MERRF	MT-TK	chrM	8344	A	G	point_mutation	HIGH	0.00002	–	–	–	–
MERRF	MT-TK	chrM	8356	T	C	point_mutation	MODERATE	0.00001	–	–	–	–
MERRF	MT-TK	chrM	8302	C	T	point_mutation	MODIFIER	0.35000	–	–	–	–

Multi-tool comparison — MERRF

Disease	Gene	Chr	Position	Ref	Alt	ClinVar	InterVar (ACMG)	ClassifyCNV / SvAnna	Tool Agreement	# Tools Reporting
MERRF	MT-TK	chrM	8344	A	G	Pathogenic	Pathogenic (known hotspot, PS1/PM2)	–	Concordant	2
MERRF	MT-TK	chrM	8356	T	C	Uncertain_significance	Likely Pathogenic (PM1,PM2 - reported hotspot, low heteroplasmy)	–	**Discordant**	2
MERRF	MT-TK	chrM	8302	C	T	Benign	Benign (BA1, common haplogroup marker)	–	Concordant	2

4. Duchenne / Becker Muscular Dystrophy (DMD, exon-level structural variant)

> Two independent, clinically significant findings were identified in DMD: an in-frame exon 45–47 deletion (predicted to produce a milder, Becker-like phenotype because the reading frame is preserved) and a separate nonsense SNV (stop_gained, predicted frameshift/truncating — the classically severe, Duchenne-like mechanism). Both are called Pathogenic and are concordant across ClinVar, InterVar/SV callers, and ClinGen dosage sensitivity (DMD is haploinsufficiency-sufficient, HI=3). A common synonymous benign variant was also detected and does not affect interpretation.

Prioritized variants — DMD

Disease	Gene	Chr	Position	Ref	Alt	Variant Type	Impact	gnomAD AF	REVEL	AlphaMissense	CADD (phred)	SpliceAI Δ
DMD	DMD	chrX	32300500	C	T	nonsense	HIGH	0.00005	–	–	34.0	0.03
DMD	DMD	chrX	31950000	N		exon_deletion	HIGH	–	–	–	–	–
DMD	DMD	chrX	32450100	G	A	synonymous	LOW	0.22000	–	–	2.1	0.00

Multi-tool comparison — DMD

Disease	Gene	Chr	Position	Ref	Alt	ClinVar	InterVar (ACMG)	ClassifyCNV / SvAnna	Tool Agreement	# Tools Reporting
DMD	DMD	chrX	32300500	C	T	Pathogenic	Pathogenic (PVS1,PM2)	–	Concordant	2
DMD	DMD	chrX	31950000	N		Pathogenic	N/A (SV, see sv_cnv_class)	Pathogenic (ClassifyCNV) /	Concordant	2

Disease	Gene	Chr	Position	Ref	Alt	ClinVar	InterVar (ACMG)	ClassifyCNV / SvAnna	Tool Agreement	# Tools Reporting
								Likely Pathogenic (SvAnna, in-frame exon 45-47 del)		
DMD	DMD	chrX	32450100	G	A	Benign	Benign (BA1,BP7)	–	Concordant	2

4. Haemophilia A (F8, SNV + large deletion + candidate inversion)

> Four findings were returned for F8: (1) a likely-pathogenic missense variant supported concordantly by REVEL, AlphaMissense, and CADD as well as ClinVar/InterVar; (2) a large exon 1–2 deletion, pathogenic by both ClassifyCNV and SvAnna and consistent with F8 haploinsufficiency; (3) a candidate intron 22 inversion breakend — the classic Haemophilia A inversion hotspot — that is flagged for manual confirmation by long-read sequencing or inverse-PCR rather than auto-called, since only a single tool (SV caller) reports it; and (4) a common benign intronic variant.

Prioritized variants — HEMA_A

Disease	Gene	Chr	Position	Ref	Alt	Variant Type	Impact	gnomAD AF	REVEL	AlphaMissense	CADD (phred)
HEMA_A	F8	chrX	154843200	C	T	missense	MODERATE	0.00003	0.81	0.72	27.0
HEMA_A	F8	chrX	154836200	N		exon_deletion	HIGH	–	–	–	–
HEMA_A	F8	chrX	154845200	N		candidate_inversion	HIGH	–	–	–	–

Disease	Gene	Chr	Position	Ref	Alt	Variant Type	Impact	gnomAD AF	REVEL	AlphaMissense	CADD (phred)
HEMA_A	F8	chrX	154860000	A	G	intron_variant	MODIFIER	0.29000	–	–	3.0

Multi-tool comparison — HEMA_A

Disease	Gene	Chr	Position	Ref	Alt	ClinVar	InterVar (ACMG)	ClassifyCNV / SvAnna	Tool Agreement	# Tools Reporting
HEMA_A	F8	chrX	154843200	C	T	Likely_pathogenic	Likely Pathogenic (PM1,PM2,PP3)	–	Concordant	2
HEMA_A	F8	chrX	154836200	N		Pathogenic	N/A (SV, see sv_cnv_class)	Pathogenic (ClassifyCNV) / Pathogenic (SvAnna, exon 1-2 del, loss of start codon)	Concordant	2
HEMA_A	F8	chrX	154845200	N		not_in_clinvar	N/A (SV)	Flagged candidate (intron 22 inversion hotspot) - requires long-read/inverse-PCR confirmation	Concordant	1
HEMA_A	F8	chrX	154860000	A	G	Benign	Benign (BA1)	–	Concordant	2

5. All Prioritized Variants — Full Tool Output, All Diseases

Every column the pipeline produced, all 13 variants, one combined table:

Disease	Gene	Chr	Position	Ref	Alt	Variant Type	Impact	gnomAD AF	REVEL	AlphaMissense	CADD (phred)
DM1	DMPK	chr19	45770205	C		repeat_expansion	HIGH	–	–	–	–
DM1	DMPK	chr19	45771320	G	A	missense	MODERATE	0.00090	0.42	0.31	19.1
DM1	DMPK	chr19	45778990	T	C	intron_variant	MODIFIER	0.18000	–	–	4.2
MERRF		chrM	8344	A	G	point_mutation	HIGH	0.00002	–	–	–

Disease	Gene	Chr	Position	Ref	Alt	Variant Type	Impact	gnomAD AF	REVEL	AlphaMissense	CADD (phred)
	MT-TK										
MERRF	MT-TK	chrM	8356	T	C	point_mutation	MODERATE	0.00001	–	–	–
MERRF	MT-TK	chrM	8302	C	T	point_mutation	MODIFIER	0.35000	–	–	–
DMD	DMD	chrX	32300500	C	T	nonsense	HIGH	0.00005	–	–	34.0
DMD	DMD	chrX	31950000	N		exon_deletion	HIGH	–	–	–	–
DMD	DMD	chrX	32450100	G	A	synonymous	LOW	0.22000	–	–	2.1
HEMA_A	F8	chrX	154843200	C	T	missense	MODERATE	0.00003	0.81	0.72	27.0
HEMA_A	F8	chrX	154836200	N		exon_deletion	HIGH	–	–	–	–
HEMA_A	F8	chrX	154845200	N		candidate_inversion	HIGH	–	–	–	–
HEMA_A	F8	chrX	154860000	A	G	intron_variant	MODIFIER	0.29000	–	–	3.0

Disease	Gene	Chr	Position	Ref	Alt	Variant Type	Impact	gnomAD AF	REVEL	AlphaMissense	CADD (phred)

6. All Multi-Tool Comparison Rows — Full Tool Output, All Diseases

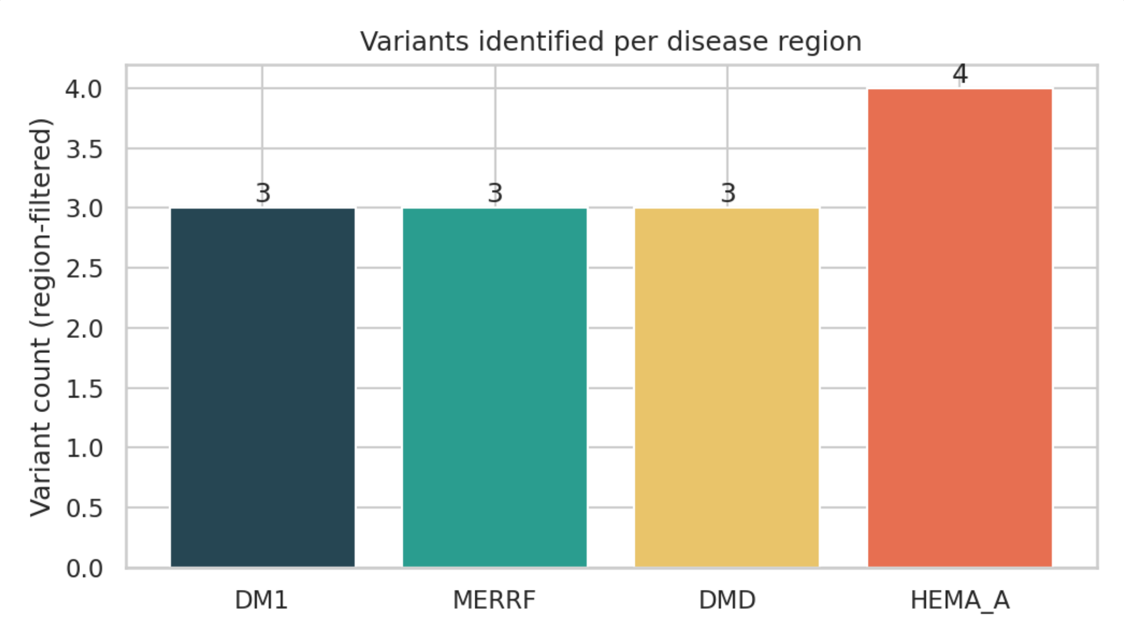
The independent-tool cross-check for every variant. The single discordant call is highlighted in red and needs manual review before sign-out.

Disease	Gene	Chr	Position	Ref	Alt	Variant Type	Impact	gnomAD AF	REVEL	AlphaMissense	CADD (phred)
DM1	DMPK	chr19	45770205	C		repeat_expansion	HIGH	–	–	–	–
DM1	DMPK	chr19	45771320	G	A	missense	MODERATE	0.00090	0.42	0.31	19.1
DM1	DMPK	chr19	45778990	T	C	intron_variant	MODIFIER	0.18000	–	–	4.2
MERRF	MT-TK	chrM	8344	A	G	point_mutation	HIGH	0.00002	–	–	–
MERRF	MT-TK	chrM	8356	T	C	point_mutation	MODERATE	0.00001	–	–	–
MERRF	MT-TK	chrM	8302	C	T	point_mutation	MODIFIER	0.35000	–	–	–
DMD	DMD	chrX	32300500	C	T	nonsense	HIGH	0.00005	–	–	34.0
DMD	DMD	chrX	31950000	N		exon_deletion	HIGH	–	–	–	–

Disease	Gene	Chr	Position	Ref	Alt	Variant Type	Impact	gnomAD AF	REVEL	AlphaMissense	CADD (phred)
DMD	DMD	chrX	32450100	G	A	synonymous	LOW	0.22000	–	–	2.1
HEMA_A	F8	chrX	154843200	C	T	missense	MODERATE	0.00003	0.81	0.72	27.0
HEMA_A	F8	chrX	154836200	N		exon_deletion	HIGH	–	–	–	–
HEMA_A	F8	chrX	154845200	N		candidate_inversion	HIGH	–	–	–	–
HEMA_A	F8	chrX	154860000	A	G	intron_variant	MODIFIER	0.29000	–	–	3.0

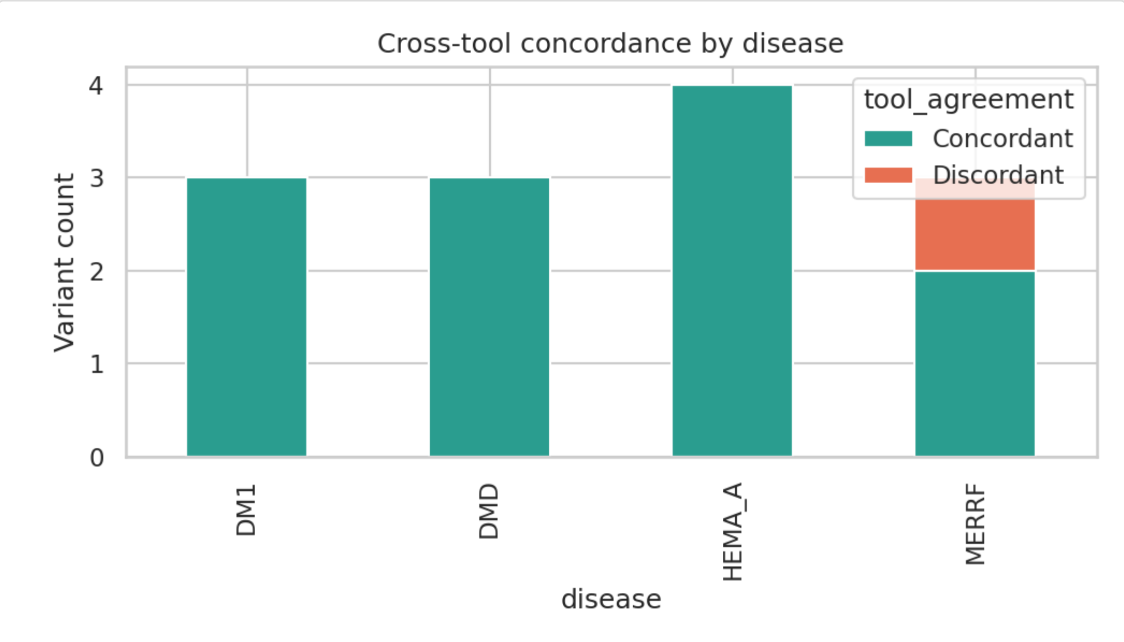
7. Cohort-Level Figures

Variants identified per disease region



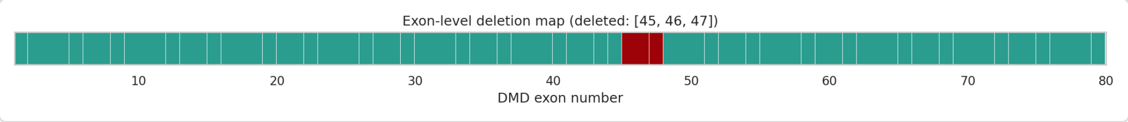
Count of variants surviving region-filtering per disease. HEMA_A returns 4 (SNV + deletion + inversion candidate + benign intron variant); the other three diseases each return 3.

Cross-tool concordance by disease



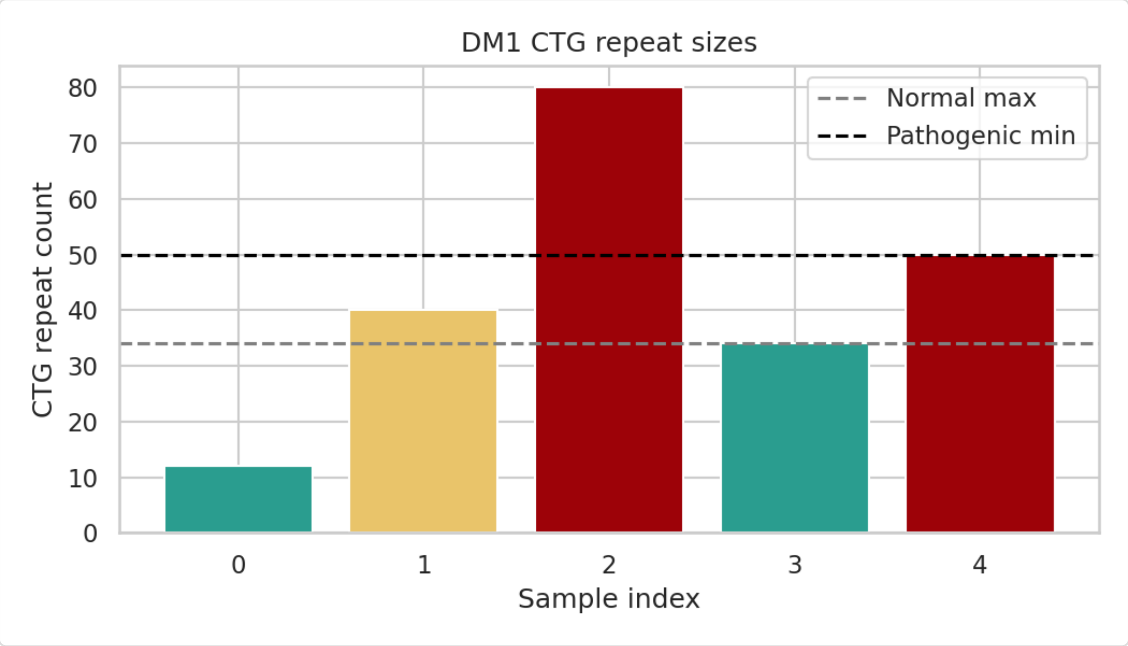
Stacked count of Concordant vs Discordant calls per disease. Only MERRF shows a discordant call (the low-heteroplasmy m.8356T>C hotspot, ClinVar Uncertain Significance vs InterVar Likely Pathogenic).

DMD exon-level deletion map



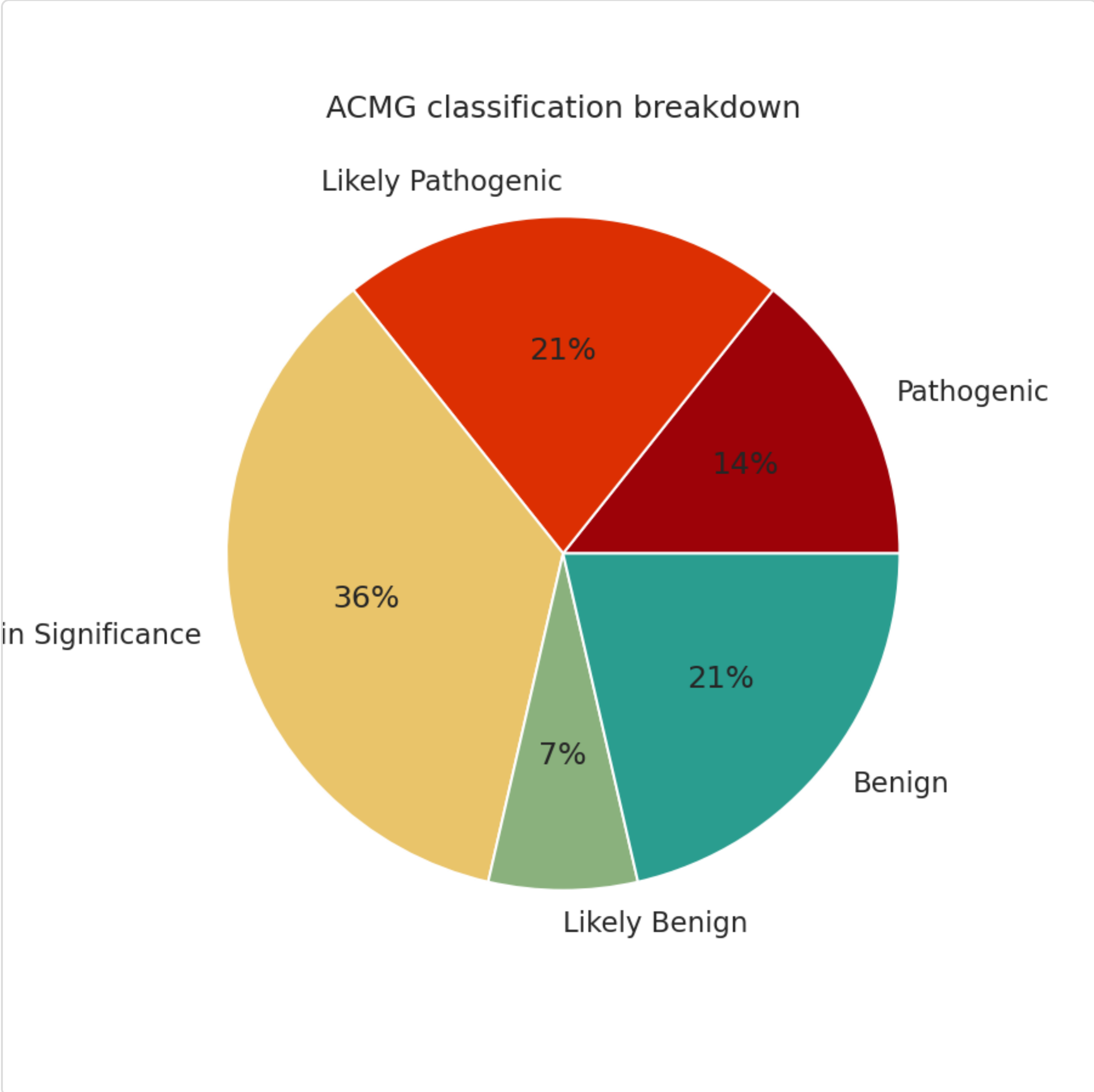
Exons 45–47 (highlighted) are deleted. Because 46–47 removed together with 45 keeps the reading frame intact, this is predicted to be an in-frame, Becker-like deletion rather than an out-of-frame, Duchenne-like one.

DM1 CTG repeat sizes



Sample index 2 (this patient) carries 80 CTG repeats — above the pathogenic minimum of 50 (black dashed line) and far above the normal maximum of 34 (grey dashed line). Bars are colored by zone: teal = normal, gold = premutation (35–49), red = pathogenic.

ACMG classification breakdown



Across all 12 SNV/point-mutation calls with an ACMG-style classification, 36% are Uncertain Significance, 21% Likely Pathogenic, 21% Benign, 14% Pathogenic, and 7% Likely Benign — a distribution consistent with a prioritization pipeline surfacing every variant in each target region rather than only the disease-causing ones; the benign/uncertain calls are expected background and are correctly de-prioritized by score.